PAN-ATLANTIC UNIVERSITY
SCHOOL OF SCIENCE AND TECHNOLOGY
Department of Computer Science

B.Sc. Computer Science **Time allowed:** 45 Minutes
2025/2026 Academic Session (Second Semester Examination)
DTS304: Data Management I
Instructions: Answer **ALL Questions**

CASE STUDY: MediBook — Clinic Management System

MediBook is a private medical clinic that wishes to digitise its day-to-day operations. The system must track doctors, patients, departmental structure, appointment scheduling, prescription records, and the clinic's medication dispensary. The following business rules govern the system:

1. The clinic is organised into **DEPARTMENTs** (e.g., Cardiology, General Practice, Dermatology). Each department has a unique department ID, a name, and a physical location within the clinic building.
2. Each **DOCTOR** is assigned to exactly one DEPARTMENT, but a DEPARTMENT may have many DOCTORs. A DOCTOR is identified by a doctor ID and also has a full name, specialisation, and phone number.
3. A **PATIENT** registers with the clinic and is assigned a unique patient ID. The system also records their full name (stored as first name and last name), date of birth, gender, and email address. The patient's age can be derived from the date of birth.
4. A PATIENT may book many APPOINTMENTs over time, and each **APPOINTMENT** involves exactly one PATIENT and one DOCTOR. Each APPOINTMENT has a unique appointment ID, date, time, and status (which must be one of: Scheduled, Completed, or Cancelled). A DOCTOR may be scheduled for many APPOINTMENTs.
5. Following a completed appointment, the attending DOCTOR may issue one or more **PRESCRIPTIONs**. A PRESCRIPTION cannot exist independently of the APPOINTMENT during which it was issued — it is therefore a weak entity. A PRESCRIPTION has a prescription number (partial key), dosage instructions, and an issue date.
6. The clinic maintains a dispensary of **MEDICATIONs**. Each MEDICATION is uniquely identified by a medication code and has a name, manufacturer, and unit price. A PRESCRIPTION may include one or more MEDICATIONs, and the same MEDICATION may appear in many PRESCRIPTIONs. The quantity prescribed for each medication in a given prescription must also be recorded.

Question 1 [15 marks]

a. Using the business rules in the case study, draw a complete **ER diagram** for the MediBook system. Your **diagram** must clearly show:

- All entities and their attributes, with primary keys correctly identified. Identify and represent the attributes accordingly in your ER Diagram based on your interpretation.
- All relationships between entities, with correct cardinality constraints.

[9 marks]

b. Convert your ER diagram from **Question 1(a)** into a complete relational database schema. Present your answer as a set of table definitions in the format below:

`Table1(`<u>`Attribute1`</u>`, Attribute2, Attribute3, …)`

`Table2(`<u>`Attribute1`</u>`, Attribute2:Table1, Attribute3, …)`

In the format above, ***Attribute2*** in ***Table2*** is a foreign key, referencing ***Table1***.

[6 marks]

Question 2 [10 marks]

a. Answer the following questions concerning ***<u>Data Models</u>***.

  i. State **the MAJOR** structural limitation of the Hierarchical Data Model, and briefly explain how the Network Model attempted to address it. **[2 marks]**

  ii. Despite the improvements offered by the Network Model over the Hierarchical Model, it was eventually replaced by the Relational Model. Identify **TWO** advantages the Relational Model offers over both early models. **[2 marks]**

b. Use **<u>your own</u>** schema in **Question 1(b)** to answer the following SQL questions:

  i. Write the **SQL CREATE TABLE** statement for the **Appointment** table based on your knowledge of Primary key, Foreign key, and other relevant associated constraints learned in class. **[3 marks]**

  ii. Write a **SELECT** query to retrieve the first name, last name, and date of birth of all female patients (Gender = 'Female'), sorted alphabetically by last name in **descending order**. **[1.5 marks]**

  iii. A new patient — Adaeze Okafor (DOB: 12th June 1998, Gender: Female, Email: adaeze.o@mail.com) — has just registered. Her assigned PatientID is 'P2091'. Write an **INSERT** statement to add her record to the Patient table. **[1.5 marks]**